
Codd's Rules (Relational Database Rules)

Proposed by **Dr. E. F. Codd**, these rules define what a **true Relational Database Management System (RDBMS)** should follow.

Rule 0: Foundation Rule

A system must manage data **entirely through relational capabilities**.

Example:

A DBMS should use tables, rows, columns, and SQL — not file-based tricks.

Rule 1: Information Rule

All data must be stored **only in tables** (relations).

Example:

A student database stores data like this:

StudentID	Name	Age
101	Alice	20

No hidden files or pointers.

Rule 2: Guaranteed Access Rule

Every data value must be accessible using:

- Table name
- Primary key
- Column name

Example:

```
SELECT Age FROM Student WHERE StudentID = 101;
```

Rule 3: Systematic Treatment of NULL Values

NULLs represent **missing or unknown data** and must be handled consistently.

Example:

If a student hasn't provided age:

StudentID	Name	Age
102	Bob	NULL

NULL $\neq$ 0 $\neq$ empty string.

Rule 4: Dynamic Online Catalog

The database structure (metadata) must be stored as tables and be queryable.

Example:

```
SELECT * FROM INFORMATION_SCHEMA.TABLES;
```

Rule 5: Comprehensive Data Sublanguage

The system must support **one complete language** for:

- Data definition
- Manipulation
- Security
- Transactions

Example:

SQL supports:

```
CREATE, INSERT, UPDATE, DELETE, GRANT, COMMIT
```

Rule 6: View Updating Rule

All views that are theoretically updatable must be updatable.

Example:

```
CREATE VIEW AdultStudents AS
```

```
SELECT * FROM Student WHERE Age >= 18;
```

You should be able to update this view.

Rule 7: High-Level Insert, Update, Delete

Operations must work on **sets of rows**, not just one row at a time.

Example:

```
UPDATE Student SET Age = Age + 1;
```

Updates all rows at once.

Rule 8: Physical Data Independence

Changes in physical storage should not affect applications.

Example:

Moving data from HDD to SSD should not break SQL queries.

Rule 9: Logical Data Independence

Changes in table structure should not affect applications.

Example:

Adding a new column `Email` should not break existing queries.

Rule 10: Integrity Independence

Integrity constraints must be defined in the database, not in application code.

Example:

```
Age INT CHECK (Age >= 0)
```

Rule 11: Distribution Independence

Users should not know where data is physically stored.

Example:

Whether data is on one server or multiple servers, SQL works the same.

Rule 12: Non-Subversion Rule

Low-level access must not bypass database rules.

Example:

No program should update data directly on disk and ignore constraints.

Quick Summary Table

Rule	Meaning
Rule 0	Must be relational
Rule 1	Data stored as tables
Rule 2	Guaranteed access
Rule 3	Proper NULL handling
Rule 4	Metadata as tables
Rule 5	One complete language
Rule 6	View updates
Rule 7	Set-level operations
Rule 8	Physical independence
Rule 9	Logical independence
Rule 10	Integrity rules in DB
Rule 11	Distribution transparency
Rule 12	No rule bypassing
